3 Propan-2-ol, $(CH_3)_2CHOH$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

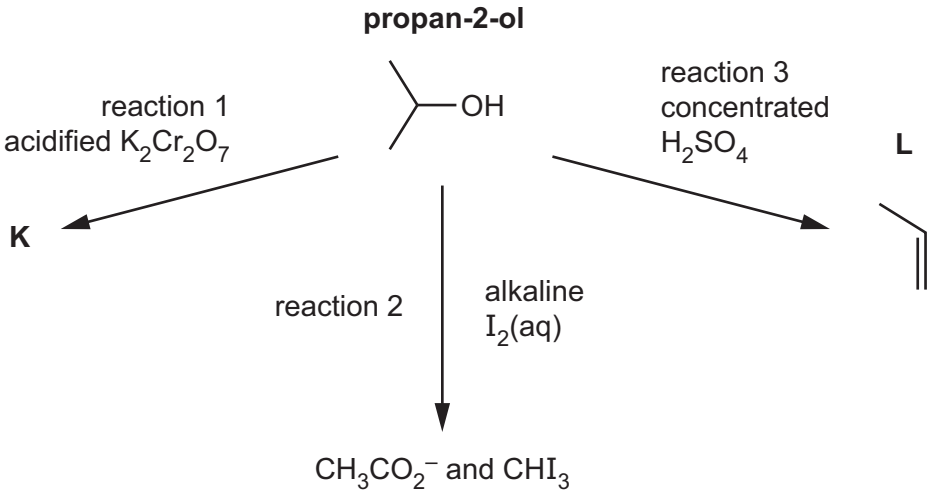


Fig. 3.1

(a) (i) Draw the structure of organic compound **K**.

[1]

(iii) State the type of reaction that is shown in reaction 3.

..... [1]

(iv) Complete Fig. 3.2 to show the pi (π) bond in **L** that is formed from orbital overlap.

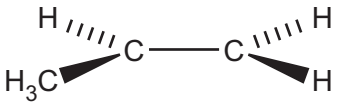


Fig. 3.2

[1]

(b) Propan-2-ol reacts with sodium to produce $(CH_3)_2CH–O^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

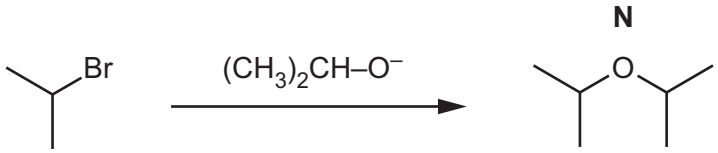


Fig. 3.3

(i) Write an equation for the reaction of propan-2-ol with sodium.

..... [1]

(ii) The reaction of 2-bromopropane with $(CH_3)_2CH–O^-$ anions follows an S_N1 mechanism.

Complete Fig. 3.4 to show this mechanism. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 3.4

[3]

4 Compounds **P** and **Q** are structural isomers.



Fig. 4.1

(a) (i) Define structural isomerism.

.....

..... [2]

(ii) **P** shows geometrical isomerism.

Draw the geometrical isomer of **P**. Explain why the two isomers are **not** identical.

geometrical isomer of **P**

explanation .....

.....

.....

[2]

(iii) Both **P** and **Q** react with aqueous bromine.

Name the mechanism of this reaction.

..... [1]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

(b) Cyclopentane, C_5H_{10}, has four cyclic structural isomers. One of these isomers is **C**, shown in Fig. 3.1.

Complete Fig. 3.1 to show two other cyclic structural isomers of C_5H_{10}.

cyclopentane			**C**
			

Fig. 3.1

[2]

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl.

(i) The reaction is initiated by the bond fission of Cl_2.

State the type of bond fission shown in the initiation step.

..... [1]

(iii) The final step is shown.

$$C_5H_9\bullet + Cl\bullet \rightarrow C_5H_9Cl$$

Give the name for this step in the reaction.

..... [1]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

[Total: 13]

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6.

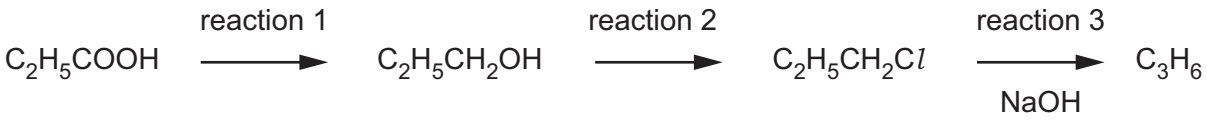


Fig. 4.1

(a) (i) Identify the type of reaction that occurs in reaction 1.

..... [1]

(iv) $C_2H_5CH_2OH$ can be directly converted to C_3H_6.

Suggest the reagent and conditions for this conversion.

..... [1]

(b) Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

(i) Identify the type of polymerisation that forms poly(propene) from propene.

..... [1]

(ii) Define stereoisomerism.

.....

..... [2]

(iii) Draw a section of poly(propene), showing **two** repeat units.

Use your diagram to identify the type of stereoisomerism shown by poly(propene). Explain your answer.

section of poly(propene)

.....

.....

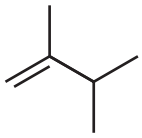
.....

.....

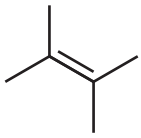
[3]

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.

X



Y



(i) Name **X**.

..... [1]

3 The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

.....

..... [2]

(c) Alkanes with low relative molecular mass, M_r, are more useful than those found in heavier crude oil fractions.

Name the process that is used to obtain alkanes with low M_r from heavier crude oil fractions.

..... [1]

(d) Hexane, C_6H_{14}, has four structural isomers.

Fig. 3.1 shows hexane and two of its structural isomers.

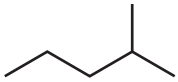
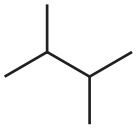
hexane	**A**	**B**	**C**	**D**
				

Fig. 3.1

(i) Complete Fig. 3.1 by drawing structures for **C** and **D**, the other two structural isomers of hexane. [2]

4 Fig. 4.1 shows how propane, C_3H_8, can be converted to propanoic acid, CH_3CH_2COOH.

$$C_3H_8 \xrightarrow{\text{reaction 1}} CH_3CH_2CH_2Cl \xrightarrow{\text{reaction 2}} CH_3CH_2CH_2OH \xrightarrow{\text{reaction 3}} CH_3CH_2COOH$$

Fig. 4.1

(a) Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.

$$C_3H_8 + Cl_2 \rightarrow CH_3CH_2CH_2Cl + HCl$$

The reaction takes place via initiation, propagation and termination steps.

(i) Name the mechanism shown by reaction 1.

..... [1]

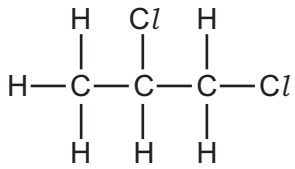
(iii) Reaction 1 is initiated by the bond fission of Cl_2.

State the type of bond fission shown in the initiation step.

..... [1]

(iv) Compound **Q** is a by-product of reaction 1.

Q



Name **Q**.

..... [1]

(v) The molecular formula of **Q** is $C_3H_6Cl_2$.

Identify the types of structural isomerism and stereoisomerism that a molecule with molecular formula $C_3H_6Cl_2$ can show.

type of structural isomerism .....

type of stereoisomerism .....

[2]

(c) Reaction 3 takes place when $CH_3CH_2CH_2OH$ is heated under reflux with acidified potassium dichromate(VI) solution.

(i) State the colour change that takes place in the reaction mixture.

..... [1]

(ii) Construct an equation to represent reaction 3. Use [O] to represent an atom of oxygen from the oxidising agent.

..... [1]

(d) CH_3CH_2COOH reacts with an unsaturated alcohol **R** to form unsaturated ester **S**.

(i) State the **type** of reaction that forms **S**.

..... [1]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

(b) Cyclopentane, C_5H_{10}, has four cyclic structural isomers. One of these isomers is **C**, shown in Fig. 3.1.

Complete Fig. 3.1 to show two other cyclic structural isomers of C_5H_{10}.

cyclopentane			**C**
			

Fig. 3.1

[2]

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl.

(i) The reaction is initiated by the bond fission of Cl_2.

State the type of bond fission shown in the initiation step.

..... [1]

(iii) The final step is shown.

$$C_5H_9\bullet + Cl\bullet \rightarrow C_5H_9Cl$$

Give the name for this step in the reaction.

..... [1]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

[Total: 13]

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6.

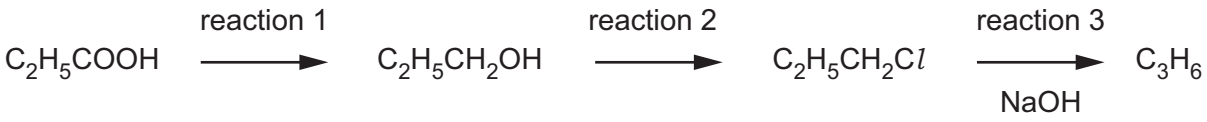


Fig. 4.1

(a) (i) Identify the type of reaction that occurs in reaction 1.

..... [1]

(iv) $C_2H_5CH_2OH$ can be directly converted to C_3H_6.

Suggest the reagent and conditions for this conversion.

..... [1]

(b) Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

(i) Identify the type of polymerisation that forms poly(propene) from propene.

..... [1]

(ii) Define stereoisomerism.

.....

..... [2]

(iii) Draw a section of poly(propene), showing **two** repeat units.

Use your diagram to identify the type of stereoisomerism shown by poly(propene). Explain your answer.

section of poly(propene)

.....

.....

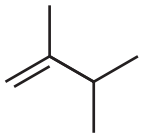
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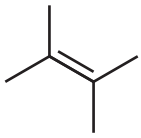
[3]

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.

X



Y



(i) Name **X**.

..... [1]

2 **(a)** Fig. 2.1 shows the covalent bonds and lone pairs of electrons in a molecule of SCl_2.

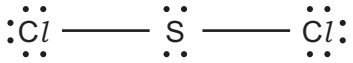


Fig. 2.1

(i) Use VSEPR theory to predict the shape of a molecule of SCl_2.

..... [1]

(ii) When SCl_2 reacts with water, the SCl_2 is broken down and a disproportionation reaction occurs.

State the oxidation numbers of sulfur in SCl_2 and H_2SO_3.

oxidation number of S in SCl_2 .....

oxidation number of S in H_2SO_3 .....
[2]

(iii) State the general name to describe the type of reaction that occurs when a substance is broken down by water.

..... [1]

4 The skeletal formula of **E** is shown in Fig. 4.1.

E

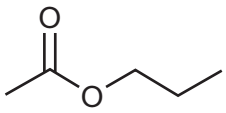


Fig. 4.1

(a) State the molecular formula of **E**.

..... [1]

(b) Name **E**.

..... [1]

(c) **E** is made when a carboxylic acid and an alcohol react together.

(i) Name the type of reaction that takes place during this esterification reaction.

..... [1]

(ii) Write an equation for the formation of **E** from a carboxylic acid and an alcohol. Use structural formulae to represent the organic species.

..... [1]

(iii) State the systematic name of the alcohol used to make **E**.

..... [1]

(d) **F** is an isomer of **E**.
F reacts with HC*l*(aq) to produce **G** and **H**.
G is a secondary alcohol.
H is also produced when HC*l*(aq) is added to CH_3CN.

(i) Construct an equation for the acid hydrolysis of CH_3CN with HC*l*(aq).

..... [1]

(ii) Draw the displayed formula of **G**.

[2]

5 The skeletal formula of **W** is shown in Fig. 5.1.

W

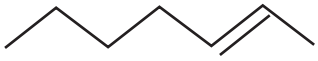


Fig. 5.1

(a) **W** has two positional isomers. Only one shows stereoisomerism.

Draw the structures of the **two** positional isomers of **W** in the boxes.

[2]

(b) **(i)** Describe the origin of stereoisomerism in **W**.

.....

..... [1]

(ii) State why **W** shows stereoisomerism but one of the positional isomers in **(a)** does **not**.

.....

.....

..... [1]

(c) Draw the skeletal structure of the stereoisomer of **W**.

[1]

6 **P**, **Q** and **R** are three different hydrocarbon molecules.

(a) (i) Define empirical formula.

.....

..... [1]

(ii) Hydrocarbon **P** contains 85.7% by mass of carbon.

Calculate the empirical formula of **P**.

Show your working.

empirical formula of **P** = ... [2]

(iii) Molecules of **P** have straight chains.

Name the homologous series to which **P** belongs.

..... [1]

5 **(a)** **W** is a colourless liquid.

W

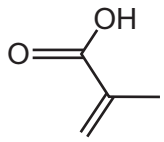


Fig. 5.1

(i) Deduce the empirical formula of **W**.

..... [1]

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.

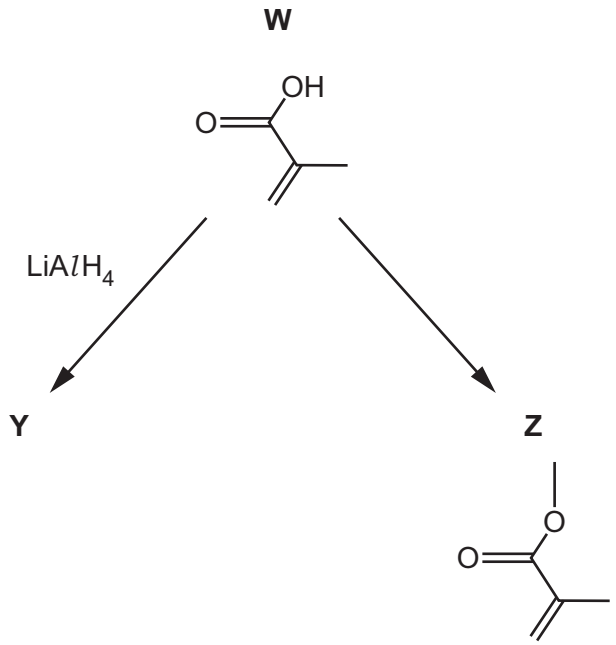


Fig. 5.2

(i) Deduce the number of sigma (σ) bonds and pi (π) bonds present in **Z**.

number of σ bonds .....

number of π bonds .....

[2]

6 **(a)** HOCl reacts with CH_2=CH_2 to produce $HOCH_2CH_2Cl$ in an electrophilic addition reaction.

(i) Define addition reaction.

.....

..... [1]

(iii) State **one** use for HOCl.

.....

..... [1]

(b) Complete Fig. 6.1 to show the mechanism for the reaction between HOCl and CH_2=CH_2 to produce $HOCH_2CH_2Cl$.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



$HO^{\delta-}$— $Cl^{\delta+}$

Fig. 6.1

[4]

(c) $HOCH_2CH_2Cl$ reacts in a two-step synthesis to produce $HOCH_2CH_2COOH$, as shown in Fig. 6.2.

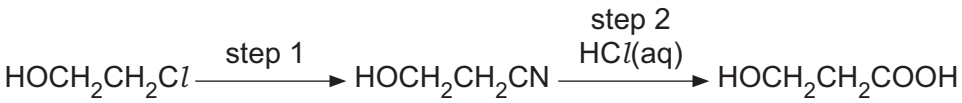


Fig. 6.2

(i) State the reagent and conditions for step 1.

..... [1]

(ii) Identify the type of reaction that occurs in step 2.

..... [1]

3 **(a)** C_3H_6 reacts with HBr(g) in an addition reaction.

(i) Define addition reaction.

.....

..... [1]

(ii) Complete Fig. 3.1 to show the mechanism for the addition reaction between C_3H_6 and HBr to produce 2-bromopropane. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 3.1

[4]

4 **(a)** Propanoic acid, CH_3CH_2COOH, reacts with reducing agent **Q** to produce propan-1-ol, $CH_3CH_2CH_2OH$.

(i) Suggest the formula of reducing agent **Q**.

..... [1]

(ii) Complete the equation to show the reduction of propanoic acid to propan-1-ol.
Use [H] to represent one atom of hydrogen from **Q**.

.....CH_3CH_2COOH + ...[H] → ..... [1]

(b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, $SOCl_2$ is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq).

(i) Name the type of reaction that occurs in step 1.

..... [1]

(iii) Construct an equation to describe the reaction of butanenitrile with NaOH(aq) in step 3.

..... [2]

(iv) Name the type of reaction that occurs in step 3.

..... [1]

(c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.

(i) Identify the type of hybridisation shown in the saturated carbon atoms of all alcohols.

..... [1]

(d) Table 4.1 shows the results of two tests on separate samples of **A**, **B** and **C**.

Table 4.1

compound	heat under reflux with acidified $K_2Cr_2O_7$	warm with alkaline I_2(aq)
A	remains orange	no visible change
B	orange to green	no visible change
C	orange to green	pale yellow precipitate

(i) Deduce the structure of **A**.

[1]

(ii) Deduce a possible name for **B**.

..... [1]

5 The structure of vitamin C is shown in Fig. 5.1.

vitamin C

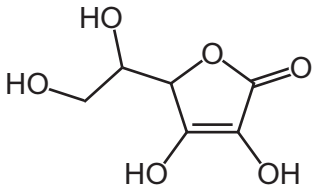


Fig. 5.1

(a) Deduce the empirical formula of vitamin C.

..... [1]

4 The structure of vitamin C is shown in Fig. 4.1.

vitamin C

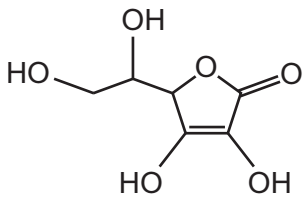


Fig. 4.1

(a) Deduce the empirical formula of vitamin C.

..... [1]

[Total: 9]

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C−O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C−H	alkane	2850–2950
N−H	amine, amide	3300–3500
O−H	carboxyl hydroxy	2500–3000 3200–3650

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $H_2SO_4(aq)$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

$$\mathbf{A} + 2H_2O \xrightarrow[\text{heat}]{H_2SO_4(aq)} \underset{\mathbf{B}}{HOOCCH_2COOH} + \underset{\mathbf{C}}{2CH_3CH_2OH}$$

Fig. 5.1

(a) Draw the structure of **A**.

[2]

(b) Name the type of reaction that occurs when **A** is heated with $H_2SO_4(aq)$.

..... [1]

(d) A student suggests a two-step synthesis to produce **B**, as shown in Fig. 5.2.

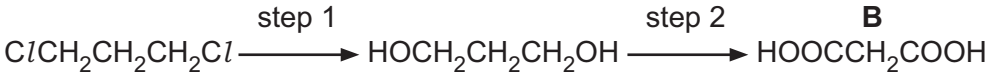


Fig. 5.2

(i) Identify the reagents and conditions required in steps 1 and 2.

step 1 .....

step 2 .....
[2]

(ii) Identify the type of reaction that $HOCH_2CH_2CH_2OH$ undergoes in step 2.

..... [1]

(e) **B** reacts with an excess of reducing agent **R** to produce **D**.

An excess of PCl_5 is added to **D**.

- A vigorous reaction occurs.
- Misty fumes are seen.
- Organic compound **E** is produced.

(i) Identify the formula of the reducing agent **R**.

..... [1]

(ii) Complete the equation to describe the reaction of **B** with an excess of **R**.
Use [H] to represent one atom of hydrogen from **R**.

.....$HOOCCH_2COOH$ + ...[H] $\rightarrow$ ..... [2]

(iii) Name organic compound **E**.

..... [1]

6 Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.

(a) State the functional group in ethanal, propanal and propanone.

..... [1]

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(a) Chlorine forms three different compounds with phosphorus.

The most common compounds are PCl_3 and PCl_5.

(i) Complete Table 1.1.

Table 1.1

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(c) 1,2-dichloroethane, $ClCH_2CH_2Cl$, reacts with NaOH to produce an unsaturated compound, C_2H_3Cl, as shown in reaction 7.

reaction 7 $ClCH_2CH_2Cl + NaOH \rightarrow C_2H_3Cl + H_2O + NaCl$

(i) State what is meant by unsaturated.

.....

..... [1]

3 The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

(a) (i) Complete Table 3.1. [1]

(d) Iodine monobromide, IBr, is a dark red solid that melts near room temperature.

IBr reacts with propene. The mechanism for this reaction is the same as the mechanism for that of HBr with propene.

(i) Identify **all** the intermolecular forces that exist between molecules of IBr.

.....

..... [1]

(ii) Name the mechanism involved in the reaction of IBr with propene.

..... [1]

(iii) The reaction of IBr with propene forms two structural isomers.

Draw the **two** structural isomers shown by these molecules.

[2]

(iv) Identify the type of structural isomerism shown by the molecules in **(d)(iii)**.

..... [1]

4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

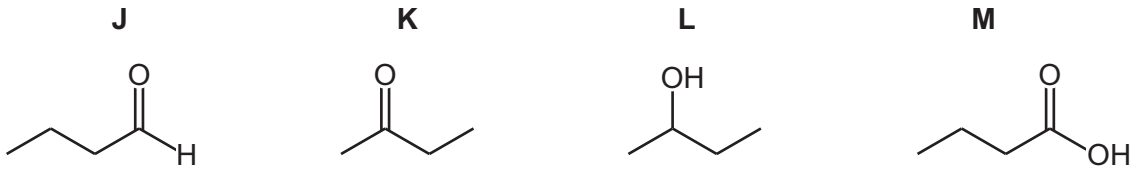


Fig. 4.1

(a) Table 4.1 gives details of tests on **J** to **M**. In each test, only **two** compounds give a positive result.

Complete Table 4.1.

Table 4.1

reagent	observation of positive result	compounds that give a positive result
acidified $K_2Cr_2O_7$(aq)		**J** and **L**
alkaline I_2(aq)	yellow precipitate	
	orange precipitate	**J** and **K**
Na(s)		

[5]

(b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

(i) Complete Fig. 4.2 to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[4]

(iii) **N** has a chiral centre.

Explain what is meant by a chiral centre.

.....

.....

..... [1]

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is [Ar]$3d^7 4s^2$.

(i) Identify what is meant by [Ar] by giving its full electronic configuration.

..... [1]

(d) Hydrated rhodium(III) chloride, $RhCl_3 \cdot xH_2O$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $RhCl_3 \cdot xH_2O$.

Show your working.

x = ...
[2]

(ii) Define stereoisomers.

.....

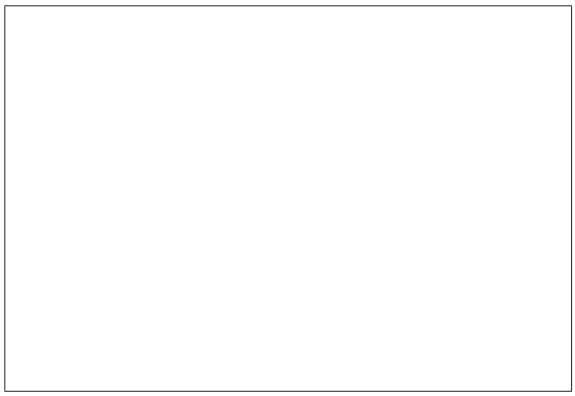
..... [1]

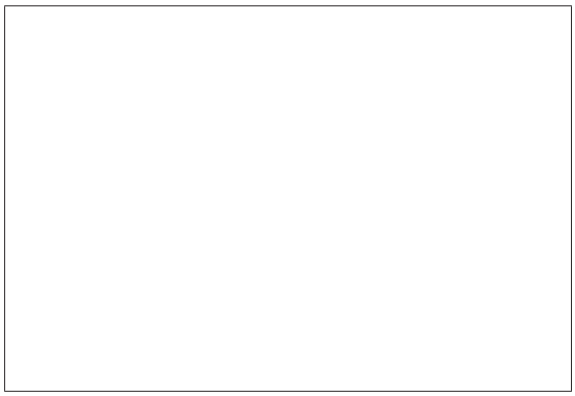
(iii) Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

.....

..... [1]

(iv) Draw the two stereoisomers of but-2-ene.





[2]

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2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) **(i)** Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(iii) State what is observed when chlorine reacts with hydrogen.

..... [1]

(iv) Explain why the thermal stability of the hydrogen halides decreases down the group.

.....

..... [1]

4 Propanone, CH_3COCH_3, is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

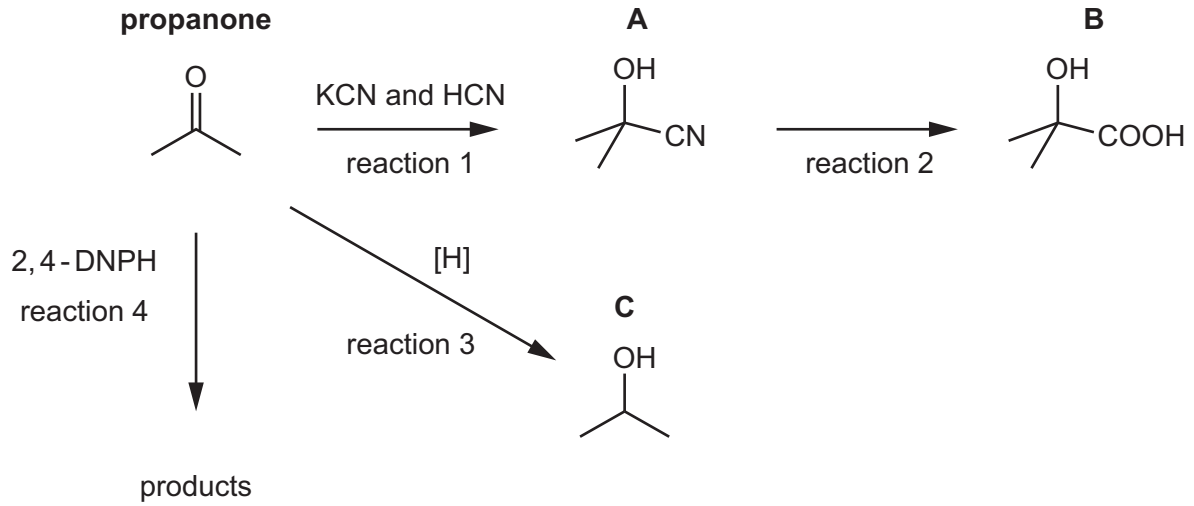


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(i) Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(ii) Explain why **A** does **not** show optical isomerism.

.....

..... [1]

(c) Reaction 3 is a reduction reaction.

(i) Construct an equation to represent reaction 3.

Use [H] to represent one atom of hydrogen from the reducing agent.

..... [1]

(ii) Name **C**.

..... [1]

4 Bromoalkanes are used widely in industry, although there is increasing concern about their environmental impact.

Fig. 4.1 shows a reaction scheme involving 1,2-dibromoethane.

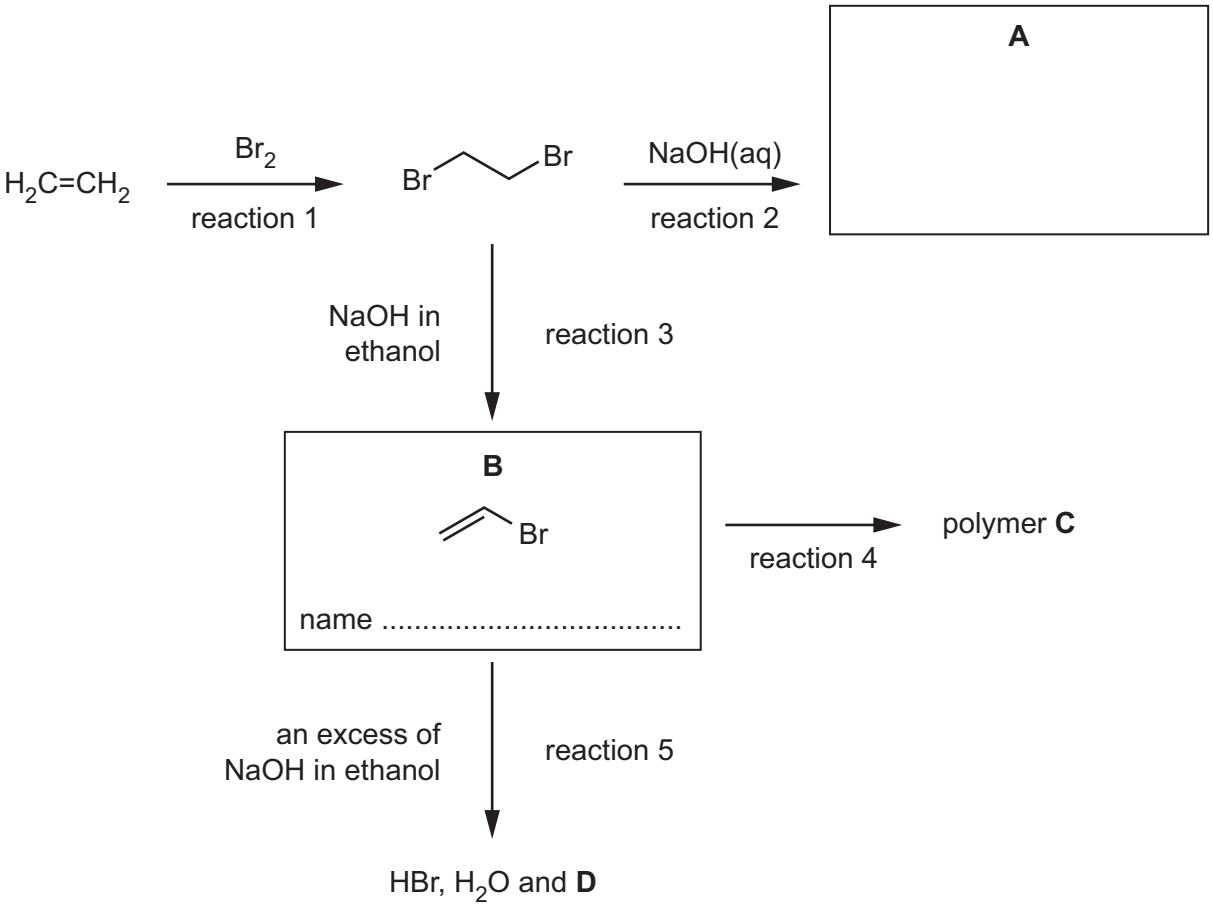


Fig. 4.1

(a) Complete Fig. 4.2 to show the mechanism for the formation of 1,2-dibromoethane in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(c) **(i)** Complete Fig. 4.1 to:

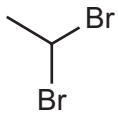
- draw the structure of compound **A**
- name compound **B**.

[2]

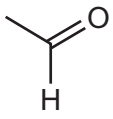
(d) Compound **E** is the only isomer of 1,2-dibromoethane.

Alkaline hydrolysis of **E** gives compound **F**.

E



F



(i) Identify the type of isomerism shown by **E** and 1,2-dibromoethane.

..... [1]

(ii) Name the homologous series that **F** belongs to.

..... [1]

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is [Ar]$3d^7 4s^2$.

(i) Identify what is meant by [Ar] by giving its full electronic configuration.

..... [1]

(d) Hydrated rhodium(III) chloride, $RhCl_3 \cdot xH_2O$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $RhCl_3 \cdot xH_2O$.

Show your working.

x = ...
[2]

(ii) Define stereoisomers.

.....

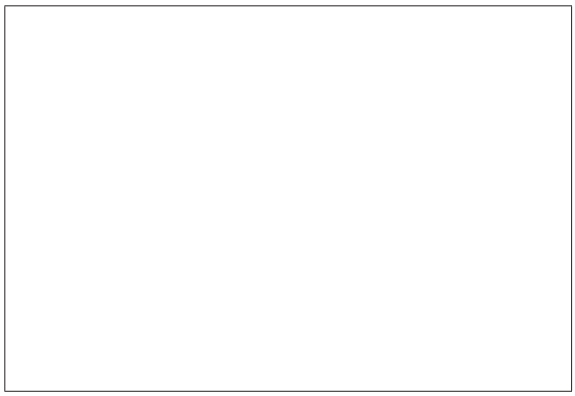
..... [1]

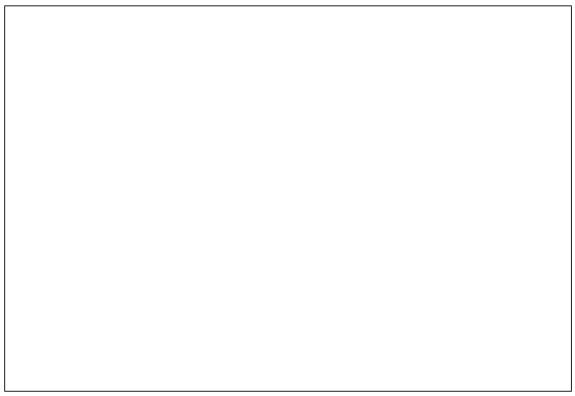
(iii) Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

.....

..... [1]

(iv) Draw the two stereoisomers of but-2-ene.





[2]

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2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) **(i)** Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(b) The halogenoalkane CH_3CH_2Cl forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.

(i) Define free radical.

.....

..... [1]

4 Propanone, CH_3COCH_3, is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

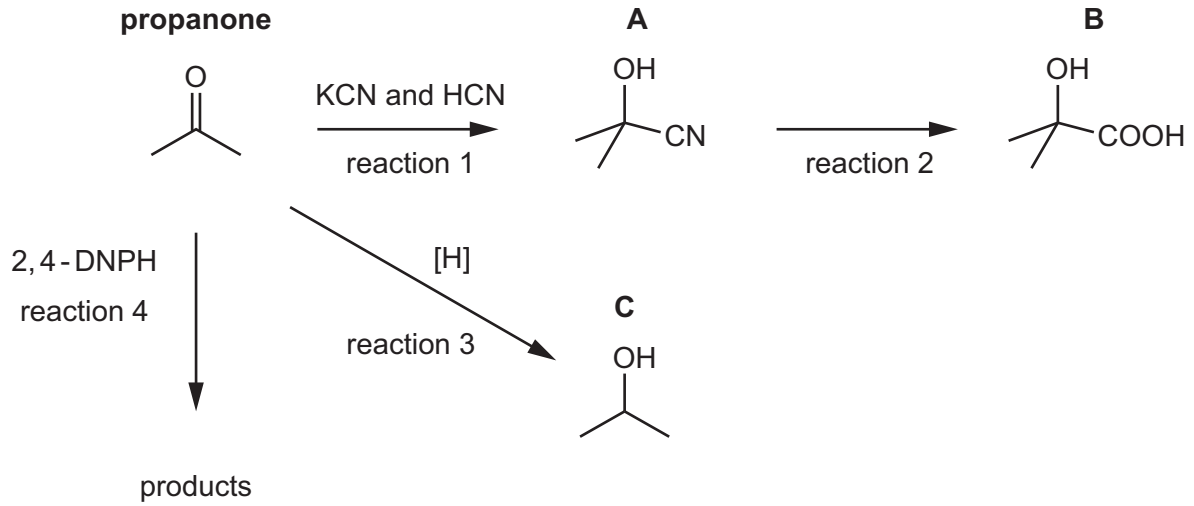


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(i) Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(ii) Explain why **A** does **not** show optical isomerism.

.....

..... [1]

(c) Reaction 3 is a reduction reaction.

(i) Construct an equation to represent reaction 3.

Use [H] to represent one atom of hydrogen from the reducing agent.

..... [1]

(ii) Name **C**.

..... [1]

4 **(a)** $NH_3(g)$ reacts with $HCl(g)$ to produce $NH_4Cl(s)$, as shown.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl.

[2]

(c) Substitution reactions of NH_3 and OH^- with halogenoalkanes both involve a lone pair of electrons.

(i) Name the role of NH_3 and OH^- in these reactions.

..... [1]

(ii) Suggest which species, NH_3 or OH^-, is more reactive during these reactions. Explain your answer.

.....

.....

..... [1]

(e) **(i)** Complete Fig. 4.2 to show the mechanism for the S_N1 reaction that occurs when $CH_3CHBrC_2H_5$ reacts with NH_3 to produce $CH_3CH(NH_2)C_2H_5$.
Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

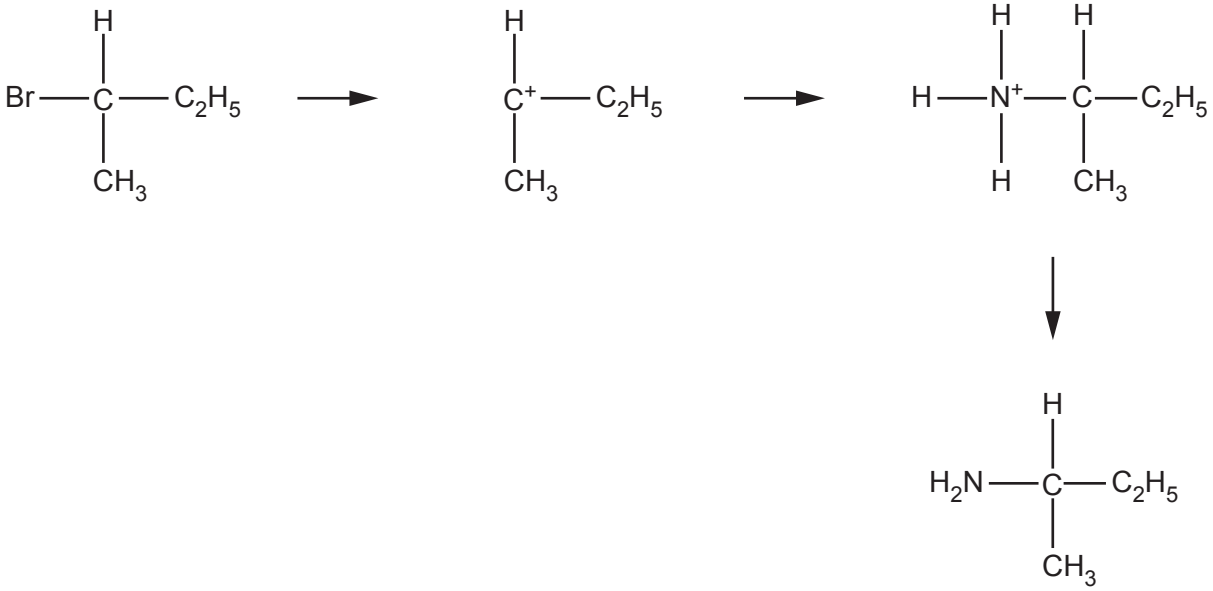


Fig. 4.2

[3]

(iii) Give the systematic name for the organic product $CH_3CH(NH_2)C_2H_5$.

..... [1]

5 **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

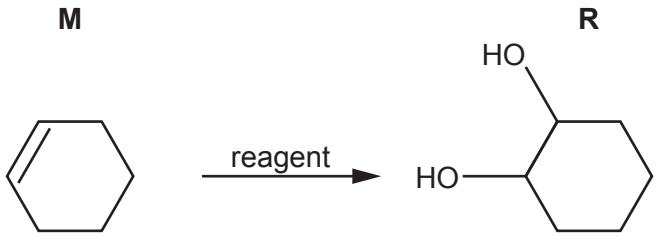


Fig. 5.1

Identify the reagent and conditions for this reaction.

..... [1]

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

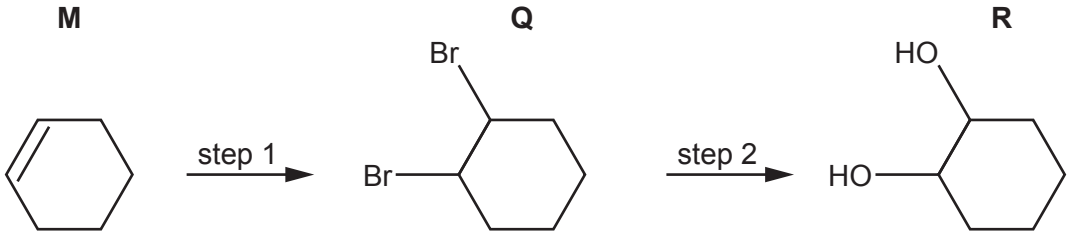


Fig. 5.2

(i) Identify the reagents and conditions for steps 1 and 2 in Fig. 5.2.

step 1 .....

step 2 .....

[2]

(ii) Name the mechanism for step 1 in Fig. 5.2.

..... [1]

(d) **Y** is made from **Q** in a three-step reaction.

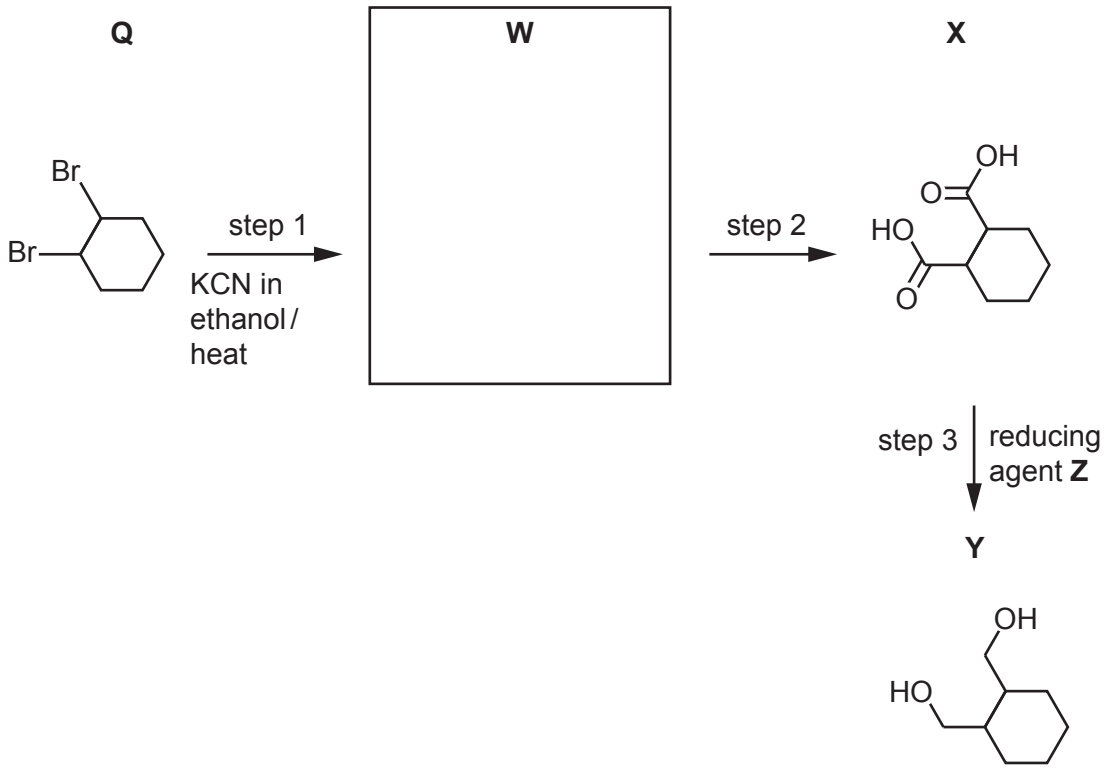


Fig. 5.4

(i) Draw the structure of **W** in the box in Fig. 5.4. [1]

(iii) In step 3, **X** reacts with reducing agent **Z** to produce **Y**.

Complete the equation for the reaction of **X** with **Z**.

Use a molecular formula to represent the organic product.

Use [H] to represent one atom of hydrogen from **Z**.

…… $C_8H_{12}O_4$ + ……[H] $\rightarrow$ ..... [1]

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4 $CH_3(CH_2)_5CHBrCH_3$ exists as a pair of stereoisomers.

(a) Draw the three-dimensional structures of the **two** stereoisomers of $CH_3(CH_2)_5CHBrCH_3$. R can be used to represent $CH_3(CH_2)_5$.

[2]

(b) A sample of $CH_3(CH_2)_5CHBrCH_3$ reacts with NaOH to make $CH_3(CH_2)_5CH(OH)CH_3$ in an S_N1 mechanism.

Complete Fig. 4.1 to show the mechanism for the reaction of $CH_3(CH_2)_5CHBrCH_3$ and NaOH.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.1

[3]

(d) $CH_3(CH_2)_5CHBrCH_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.

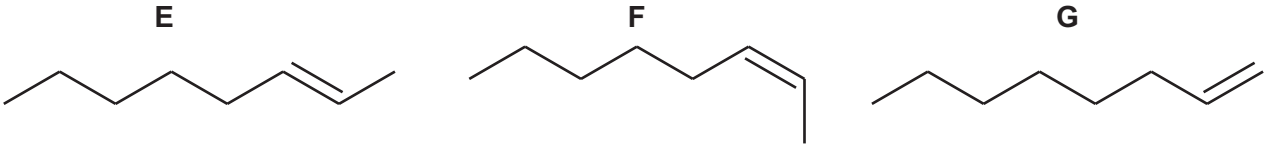


Fig. 4.2

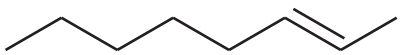
(i) Name the type of reaction.

..... [1]

(e) (i) Both σ and π bonds are present in a molecule of **E** as a result of different types of hybridisation in the carbon atoms.

Complete Table 4.2 to show the number of carbon atoms with each type of hybridisation in a molecule of **E**.

Table 4.2

	number of carbon atoms		
	sp hybridised	sp^2 hybridised	sp^3 hybridised
E 			

[2]

(ii) Describe the essential feature of an unbranched hydrocarbon that causes its molecules to show stereoisomerism. Explain how this feature leads to stereoisomerism.

.....

.....

..... [3]

5 Compound **W** has molecular formula $C_4H_{10}O$. It contains only **one** functional group.

(a) Table 5.1 shows the two peaks with the greatest *m*/*e* values in the mass spectrum of **W**.

Table 5.1

m/*e*	relative abundance
74	50
75	x

(i) Calculate the relative abundance, x, of the peak at *m*/*e* = 75 using the information from Table 5.1.

x = ... [1]

(c) **Y** is a structural isomer of **W**.

Both **W** and **Y** produce colourless bubbles when sodium is added to them.

Y does **not** react when heated with acidified $K_2Cr_2O_7$.

Y does **not** react when warmed with alkaline I_2(aq).

(i) Name the functional group present in **Y**.

..... [1]

(iii) Draw the structure of **Y**.

[1]

5 Hydrocarbon molecules contain covalent bonds.

(a) Define covalent bond.

.....

..... [1]

(b) A C=C bond in an alkene is made from a σ bond and a π bond.

(i) Identify the hybridisation of the carbon atoms in a C=C bond in an alkene.

..... [1]

(ii) Draw labelled diagrams to show, in terms of orbital overlap, how the σ and π bonds are made in a C=C bond.

σ bond

π bond

[2]

(ii) Complete Fig. 5.1 to show the mechanism for the electrophilic addition of hydrogen bromide to 2-methylpropene to produce the major organic product.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 5.1

[4]

6 **(a)** **V** shows stereoisomerism.

V

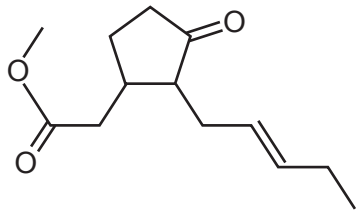


Fig. 6.1

(i) Explain what is meant by stereoisomerism.

.....

.....

..... [1]

(ii) Deduce the number of stereoisomers of **V**. Explain your reasoning.

.....

.....

..... [2]

(iii) Deduce the molecular formula of **V**.

..... [1]

(iv) Name **all** the functional groups present in **V**.

.....

..... [1]

(b) Fig. 6.2 shows two reactions involving **V**.

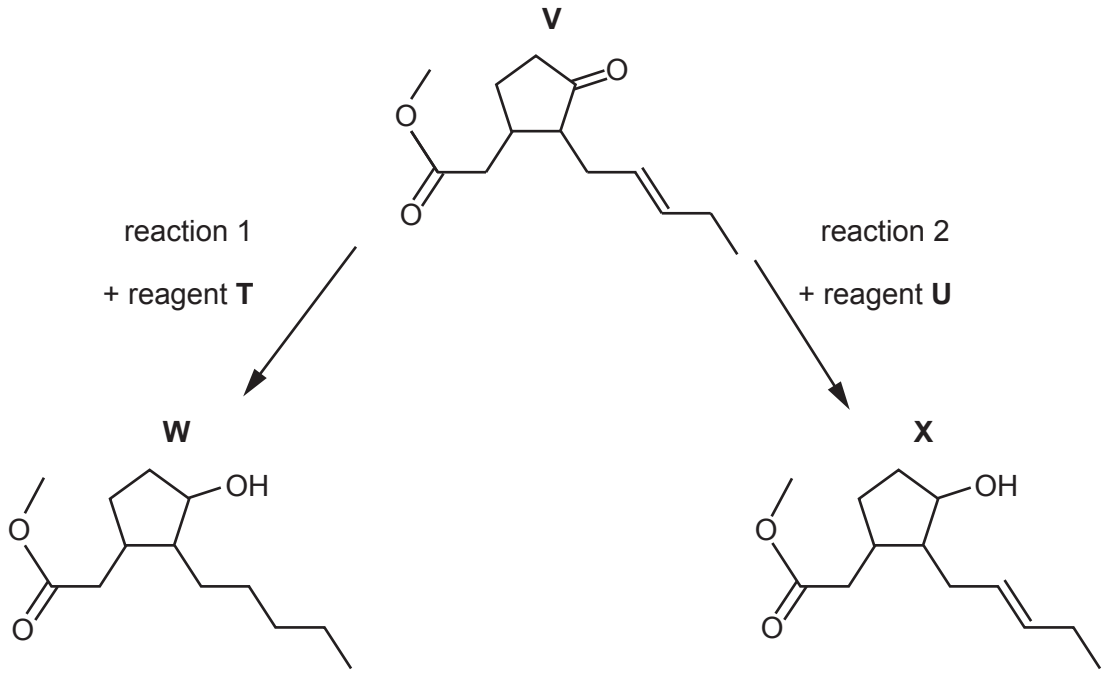


Fig. 6.2

(i) Identify the role of reagent **T** for each functional group that reacts in reaction 1.

.....

..... [1]

2 Chlorine, Cl_2, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(a) Complete Table 2.1. [3]

(c) An excess of Cl_2 reacts with phosphorus to form PCl_5.

(i) PCl_5 is a simple molecule in the gas phase.

It also exists in a solid form as two ions, PCl_4^+ and PCl_6^-.

Complete Table 2.2 to identify the shapes of each of these species.

Table 2.2

species	PCl_5	PCl_4^+	PCl_6^-
shape		tetrahedral	

[2]

(ii) PCl_5 reacts with **J** to form H_3PO_4.

Identify **J** and state the type of reaction.

J ..... type of reaction .....

[2]

(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

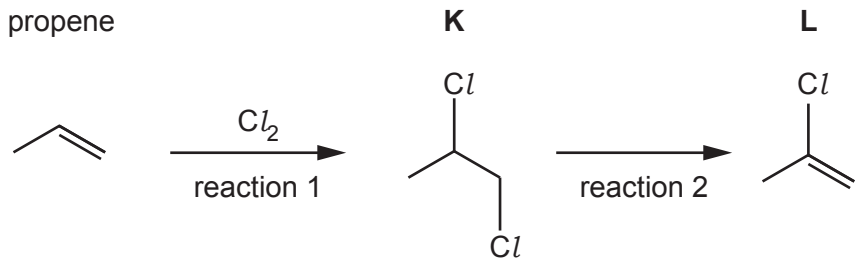


Fig. 2.1

(i) Complete Fig. 2.2 to show the mechanism for the reaction of Cl_2 with propene in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

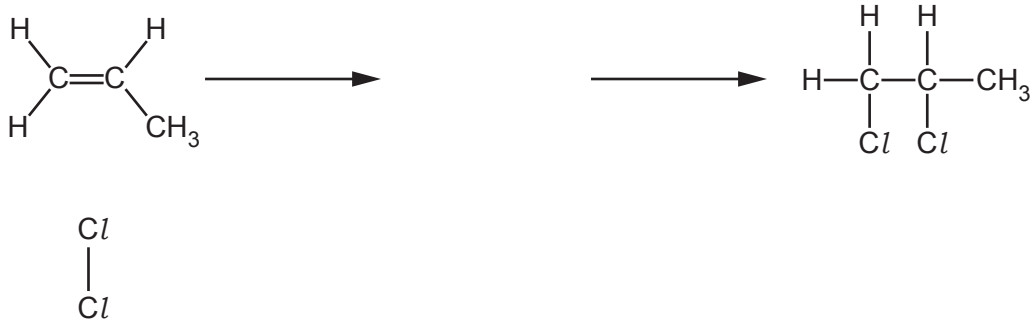


Fig. 2.2 [4]

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

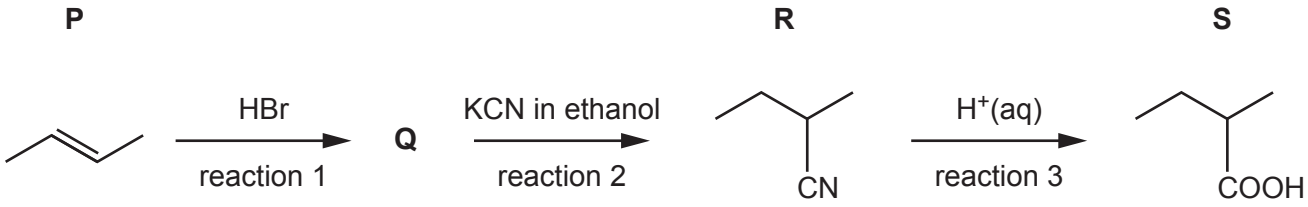


Fig. 4.1

(a) **P**, **Q**, **R** and **S** show stereoisomerism.

Complete Table 4.1 by identifying with a tick (✓) the type of stereoisomerism that each molecule shows.

The type of stereoisomerism shown by **Q** is given.

Table 4.1

	P	**Q**	**R**	**S**
geometrical isomerism				
optical isomerism		✓		

[2]

(b) **(i)** Give the structural formula of **Q**.

..... [1]

(ii) Name the mechanism in reaction 2.

..... [1]

(c) Compounds **S** and **T** react to form organic compound **U**, which has a single functional group.

S

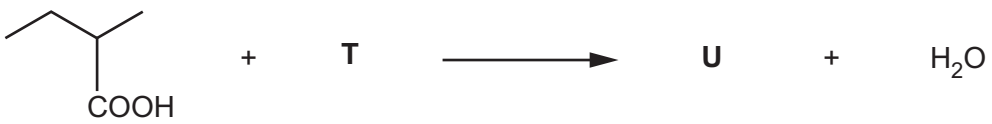


Table 4.2 shows some data from the mass spectrum of **U**.

Table 4.2

peak	relative abundance
M^+	7.2
$[M+1]^+$	0.55

(i) Use the data from Table 4.2 to show that **U** contains 7 carbon atoms.

Show your working.

[2]

(ii) Fig. 4.2 shows the infrared spectrum of **U**.

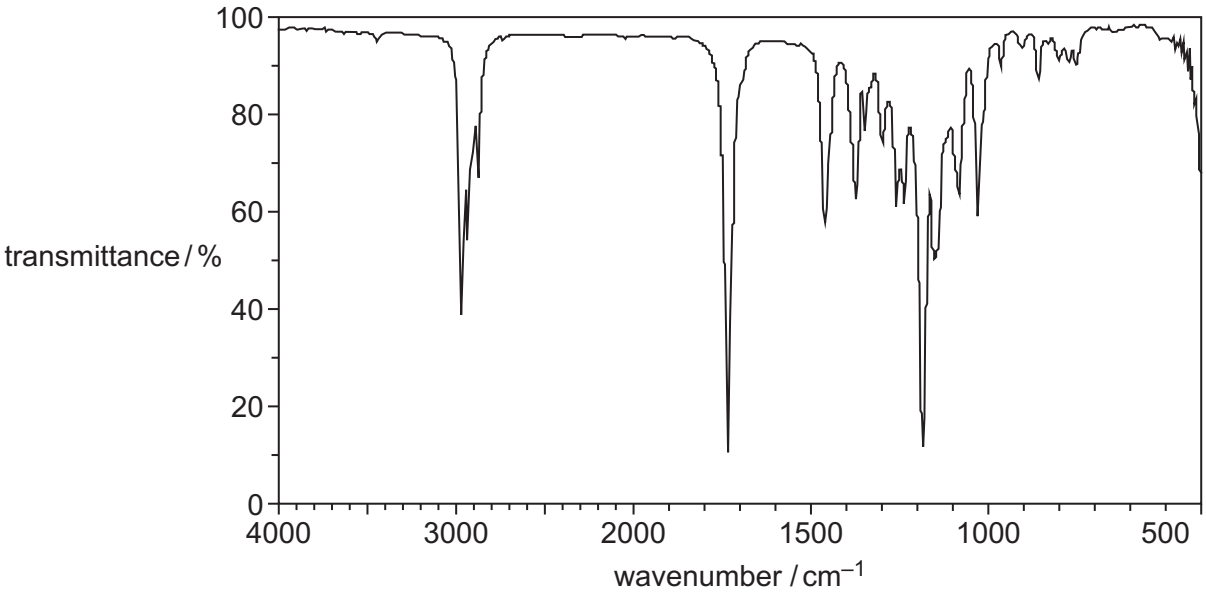


Fig. 4.2

Table 4.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Use Fig. 4.2 and Table 4.3 to identify the functional group present in **U**.

Explain your answer fully.

functional group .....

explanation .....

.....

.....

.....

[2]

(iii) **T** also has a single functional group.

Use the information in **(c)(i)** and your answer to **(c)(ii)** to identify **T** and **U**.

Draw the structures of **T** and **U** in the boxes.

T

U

[2]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(iii) There are many structural isomers of $C_6H_{12}O_6$.

Define structural isomers.

.....

..... [1]

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10}.



Fig. 4.1

(a) **(i)** Give the systematic name of **D**.

..... [1]

(ii) Explain why **C** and **D** do not show geometrical (*cis/trans*) isomerism.

.....

..... [1]

(iii) Draw the structure of a molecule that is a positional isomer of **C** and **D**.

[1]

(c) **C** and **D** both react with HBr.

(i) **C** reacts with HBr to form **E**.

Complete the diagram in Fig. 4.2 to show the mechanism for this reaction.

Draw the structure of the organic intermediate.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[3]

(ii) **D** reacts with HBr to produce **F**, a chiral bromoalkane.

Draw the structure of **F**.

F

[1]

3 Phosphoric(V) acid, H_3PO_4, is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4.

(i) Write an equation for each step.

step 1 .....

step 2 .....

[2]

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

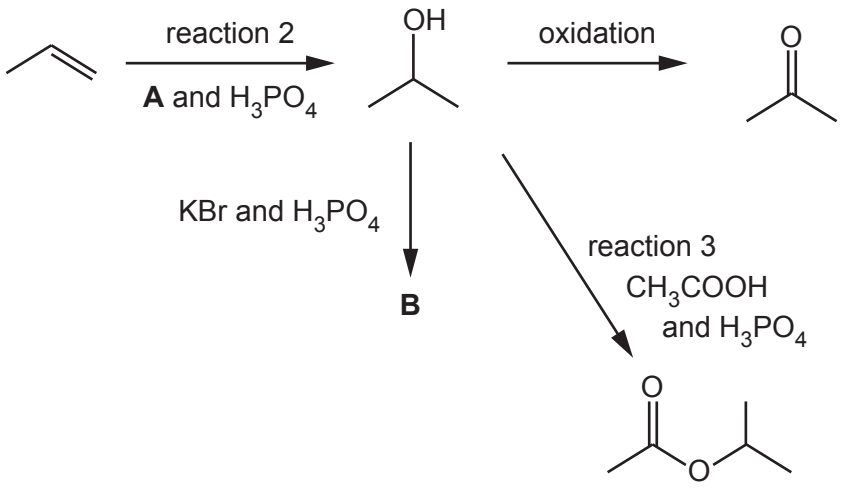


Fig. 3.1

(i) Identify **A**, which reacts with propene in the presence of H_3PO_4 in reaction 2.

..... [1]

(iii) Name the type of reaction that occurs in reaction 3.

..... [1]

4 Lactic acid, $CH_3CH(OH)COOH$, and pyruvic acid, $CH_3COCOOH$, both contain two functional groups.

lactic acid **pyruvic acid**

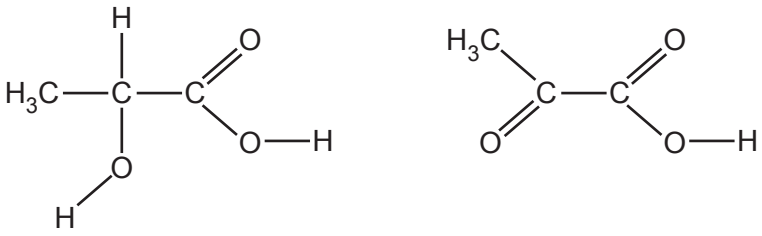


Fig. 4.1

(a) (i) Explain why lactic acid exists as optical isomers.

.....

.....

..... [1]

(ii) Give the systematic name of lactic acid.

..... [1]

(b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

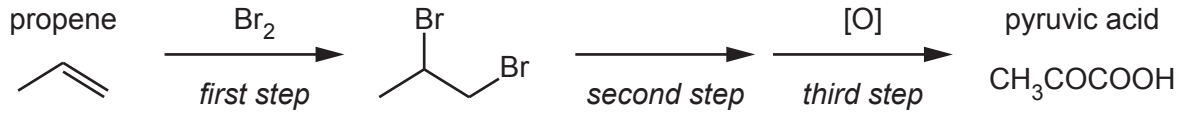


Fig. 4.3

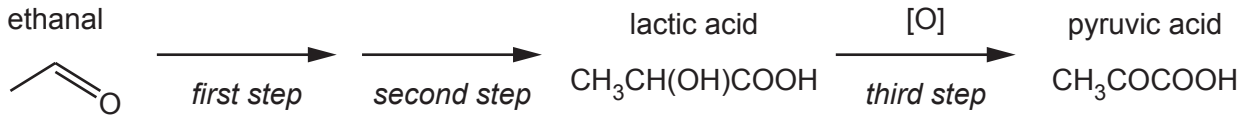


Fig. 4.4

(i) Complete the diagram in Fig. 4.5 to show the mechanism for the reaction of propene with Br_2.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.5

[3]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(iii) There are many structural isomers of $C_6H_{12}O_6$.

Define structural isomers.

.....

..... [1]

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10}.



Fig. 4.1

(a) (i) Give the systematic name of **D**.

..... [1]

(ii) Explain why **C** and **D** do not show geometrical (*cis/trans*) isomerism.

.....

..... [1]

(iii) Draw the structure of a molecule that is a positional isomer of **C** and **D**.

[1]

(c) **C** and **D** both react with HBr.

(i) **C** reacts with HBr to form **E**.

Complete the diagram in Fig. 4.2 to show the mechanism for this reaction.

Draw the structure of the organic intermediate.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[3]

(ii) **D** reacts with HBr to produce **F**, a chiral bromoalkane.

Draw the structure of **F**.

F

[1]

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.



Fig. 5.1

(a) Deduce the empirical formula of **Y**.

..... [1]

(b) The formation of **Y** from **X** requires the addition of a suitable reducing agent.

(i) Construct an equation using molecular formulae and [H] for the reaction in Fig. 5.1. Use [H] to represent one atom of hydrogen from the reducing agent.

..... [1]

(c) Complete Table 5.1 to show the number of sp^2 and sp^3 hybridised carbon atoms in a molecule of **X**.

Table 5.1

type of hybridisation	sp^2	sp^3
number of carbon atoms in **X**		

[2]

6 Compound **W**, CH_2=CHCN, is used to make an addition polymer which is present in carbon fibres.

(a) Draw **one** repeat unit of the addition polymer of **W**.

[1]

(b) CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN.

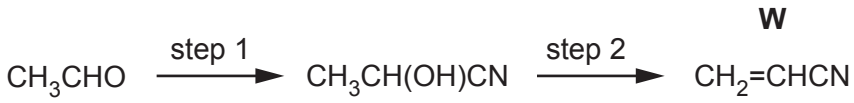


Fig. 6.1

(i) Name the mechanism for the reaction in step 1 in Fig. 6.1.

..... [1]

(ii) Complete Fig. 6.2 to show the mechanism for the reaction in step 1.
Include all products, charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

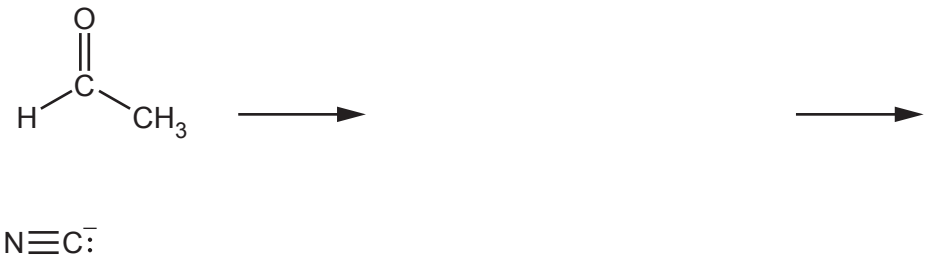


Fig. 6.2

[3]

(c) Molecules of **W**, CH_2=CHCN, do **not** show stereoisomerism.

(i) Describe stereoisomerism.

.....

..... [1]

(ii) Describe the **two** essential features of an alkene molecule that cause it to show geometrical stereoisomerism.

.....

..... [2]

(d) Molecules of $CH_3CH(OH)CN$ exist as a pair of optical isomers.

Draw three-dimensional diagrams in the boxes to show the optical isomers of $CH_3CH(OH)CN$.

isomer 1	isomer 2

[1]

(e) Propanenitrile is heated with hydrogen gas and a platinum catalyst. The only product is propylamine.

Construct an equation for this reaction.

..... [1]

(f) Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

$$CH_3CH_2CH_2OH \xrightarrow[SOCl_2]{\text{step 1}} CH_3CH_2CH_2Cl \xrightarrow{\text{step 2}} CH_3CH_2CH_2NH_2$$

Fig. 6.4

(i) Name the type of reaction in step 1 in Fig. 6.4.

..... [1]

3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from CH_3CH_2Cl.

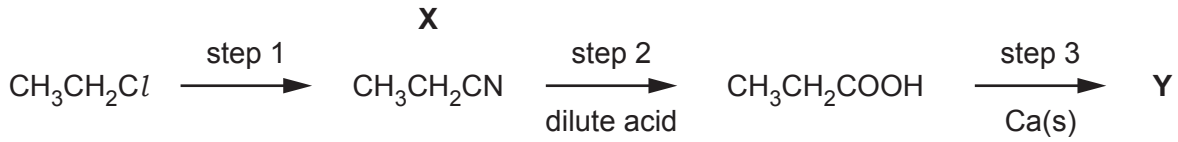


Fig. 3.1

(a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

..... [1]

(ii) Give the systematic name of **X**.

..... [1]

(iii) Identify the type of reaction that occurs when dilute acid is added to **X** in step 2.

..... [1]

(c) Complete Table 3.1 to show the number of sigma bonds (σ) and pi bonds (π) present in a molecule of **X**.

Table 3.1

type of bond	number of bonds in **X**
sigma (σ)	
pi (π)	

[2]

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4.

reaction 1 $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$

(a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r, for reaction 1.

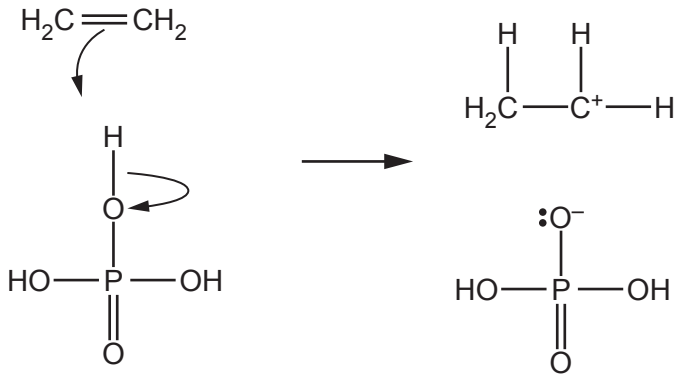
Table 4.1

bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

ΔH_r = ... kJ mol^{-1} [2]

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

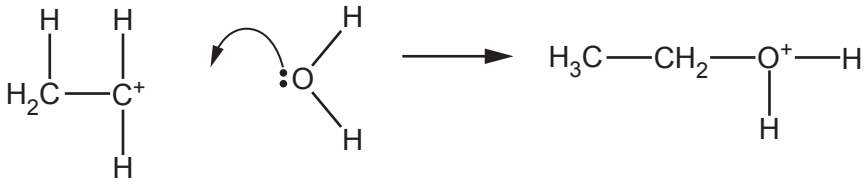


Fig. 4.1

(i) Describe the behaviour of H_3PO_4 in step 1 in Fig. 4.1. Explain your answer.

.....

..... [1]

(ii) Identify the species that behaves as an electrophile in step 2 in Fig. 4.1. Explain your answer.

.....

..... [1]

(iii) Complete Fig. 4.2 to show the mechanism for step 3 of reaction 1. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

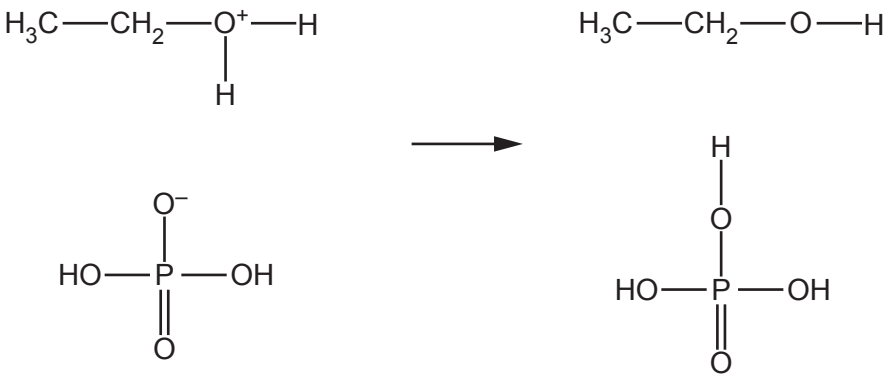


Fig. 4.2

[2]

5 **(a)** Describe structural isomerism.

.....

..... [1]

(b) **A** and **B** are structural isomers with molecular formula $C_5H_{10}O$.

They are both straight-chained molecules with only one functional group.

Table 5.1 describes observations when separate samples of **A** and **B** are added to different reagents.

Table 5.1

reagent	**A**	**B**
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	orange precipitate appears	orange precipitate appears
Tollens' reagent	silver mirror appears	no reaction
alkaline I_2(aq)	no reaction	no reaction

(i) Name the functional group present in both **A** and **B**.

..... [1]

(ii) Draw the structures of **A** and **B** in the boxes.

A	**B**

[2]

(c) **C** is a structural isomer of **A** and **B**.

C is straight chained and has two functional groups.

C shows only one type of stereoisomerism.

Table 5.2 describes observations when separate samples of **C** are added to different reagents.

Table 5.2

reagent	**C**
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	no reaction
$Br_2(aq)$	orange to colourless
alkaline $I_2(aq)$	yellow precipitate appears

(i) Draw the structure of **C** in the box.

C

[2]

(ii) Name the type of stereoisomerism shown by molecules of **C**.

..... [1]

(d) **D** reacts in the presence of a sulfuric acid catalyst to form **E** and water.

The structure of **E** is shown in Fig. 5.1.

E

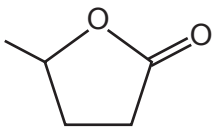


Fig. 5.1

(i) Name the functional group present in **E**.

..... [1]

(ii) Identify the type of reaction that occurs when **D** reacts to form **E**.

..... [1]

(iii) Draw the structure of **D** in the box.

D

[1]

1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te. Its electronic configuration is [Kr] $4d^{10}\ 5s^2\ 5p^4$.

(a) Complete Table 1.1.

Table 1.1

	nucleon number	number of neutrons	number of electrons
^{130}Te			

[3]

(f) TeF_x reacts with water to form tellurium hydroxide and HF. The oxidation number of tellurium does **not** change during this reaction.

(i) Construct an equation for the reaction of TeF_x with water.

..... [1]

(ii) Name the type of reaction that occurs when TeF_x reacts with water.

..... [1]

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance .....

structure and bonding .....
[2]

(d) Chlorine reacts with methane in a series of reactions to produce chloroalkanes.

(i) State the conditions required for chlorine to react with methane.

..... [1]

(ii) One of the products of the reaction is CH_2Cl_2 which reacts further to produce $CHCl_3$.

Complete Table 3.2 to show details of the mechanism that forms $CHCl_3$ from CH_2Cl_2.

Table 3.2

name of step	equation
initiation	.....
propagation	$CH_2Cl_2 + Cl\bullet \rightarrow$ .....
termination	..... $\rightarrow CHCl_3$

[3]

4 **V** is a colourless liquid.

V

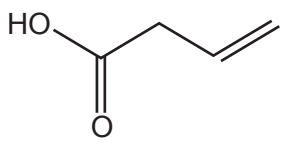


Fig. 4.1

(a) **V** reacts with an excess of $LiAlH_4$ to form **W**.

(i) Draw the structure of **W** in the box.

W

[1]

(ii) Identify the role of $LiAlH_4$ in the reaction with **V**.

..... [1]

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.

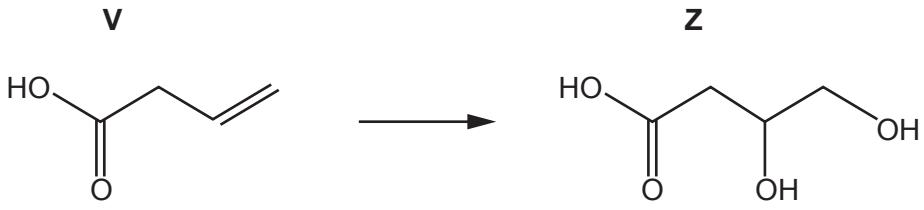


Fig. 4.2

(i) Suggest the reagent and conditions needed to form **Z** from **V**.

..... [1]

(ii) Deduce the empirical formula of **Z**.

..... [1]

(iii) Complete Table 4.1 to show the number of sp^2 and sp^3 hybridised carbon atoms that are present in a molecule of **V**.

Table 4.1

type of hybridisation	sp^2	sp^3
number of carbon atoms in **V**		

[2]

5 **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

(i) Construct an equation for the complete combustion of **M**.

..... [2]

(c) Biodiesel **T** is a fuel made from vegetable oil **R**. Fig. 5.1 shows the production of **T** from **R** in a two-step process.

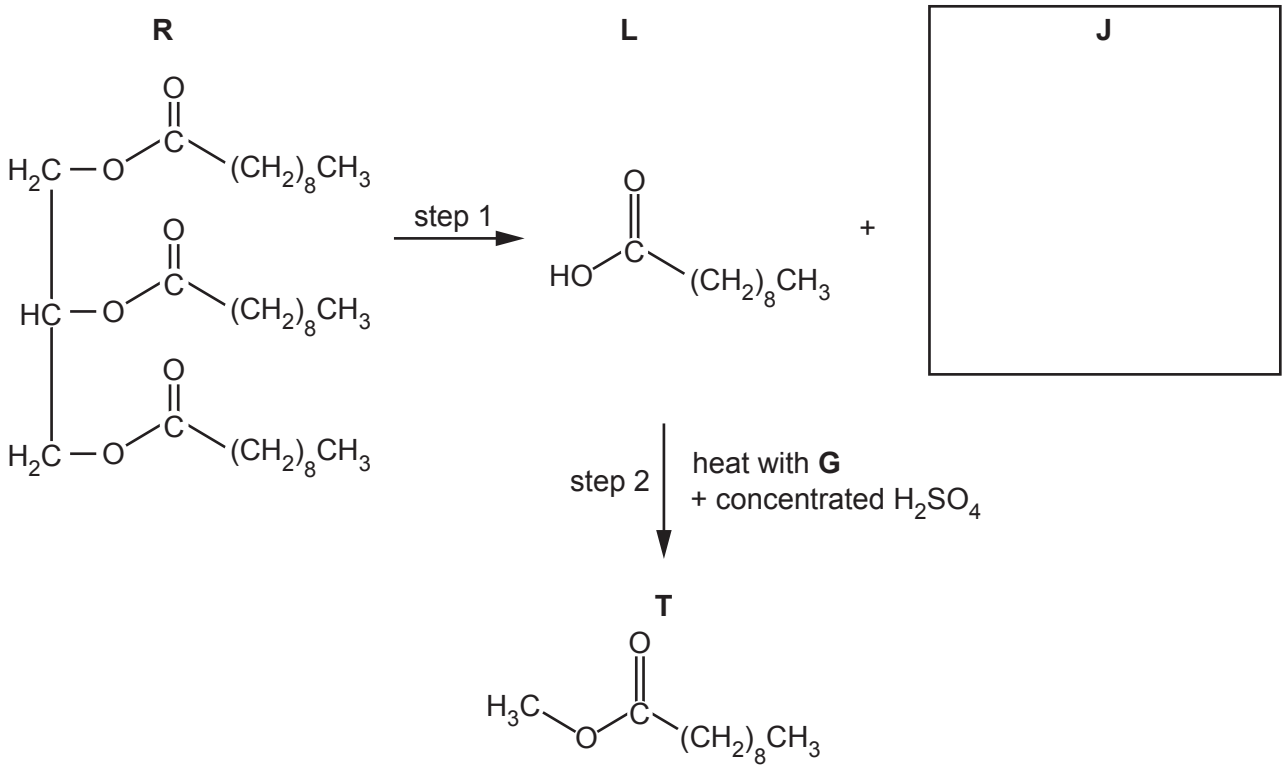


Fig. 5.1

(i) In step 1 all three ester groups in **R** react. Suggest a suitable reagent and conditions for step 1.

..... [1]

(ii) Draw the structural formula of **J** in the box in Fig. 5.1. [1]

(iii) Name the type of reaction that occurs in step 2.

..... [1]

(iv) Name organic reagent **G** used in step 2.

..... [1]

(v) **L** is called decanoic acid. Use systematic nomenclature to deduce the name of **T**.

..... [1]

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2.

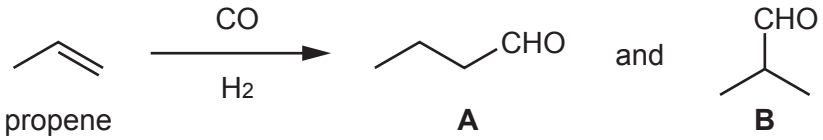


Fig. 3.1

(a) (i) Define addition reaction.

.....

..... [1]

(ii) Aldehydes **A** and **B** are structural isomers.

State the type of structural isomerism shown by **A** and **B**.

..... [1]

(iii) Name **A**.

..... [1]

(b) **A** and **B** show reactions typical of aliphatic aldehydes.

(i) **A** undergoes a nucleophilic addition reaction with a mixture of HCN and KCN, forming compound **C**.

Complete the diagram to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

Draw the structure of the organic intermediate.

A



C

[4]

(iv) **C** is a chiral molecule.

Circle any chiral centres in the structure of **C** shown in Fig. 3.2.

C

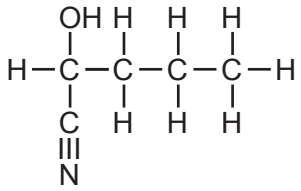


Fig. 3.2

[1]

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

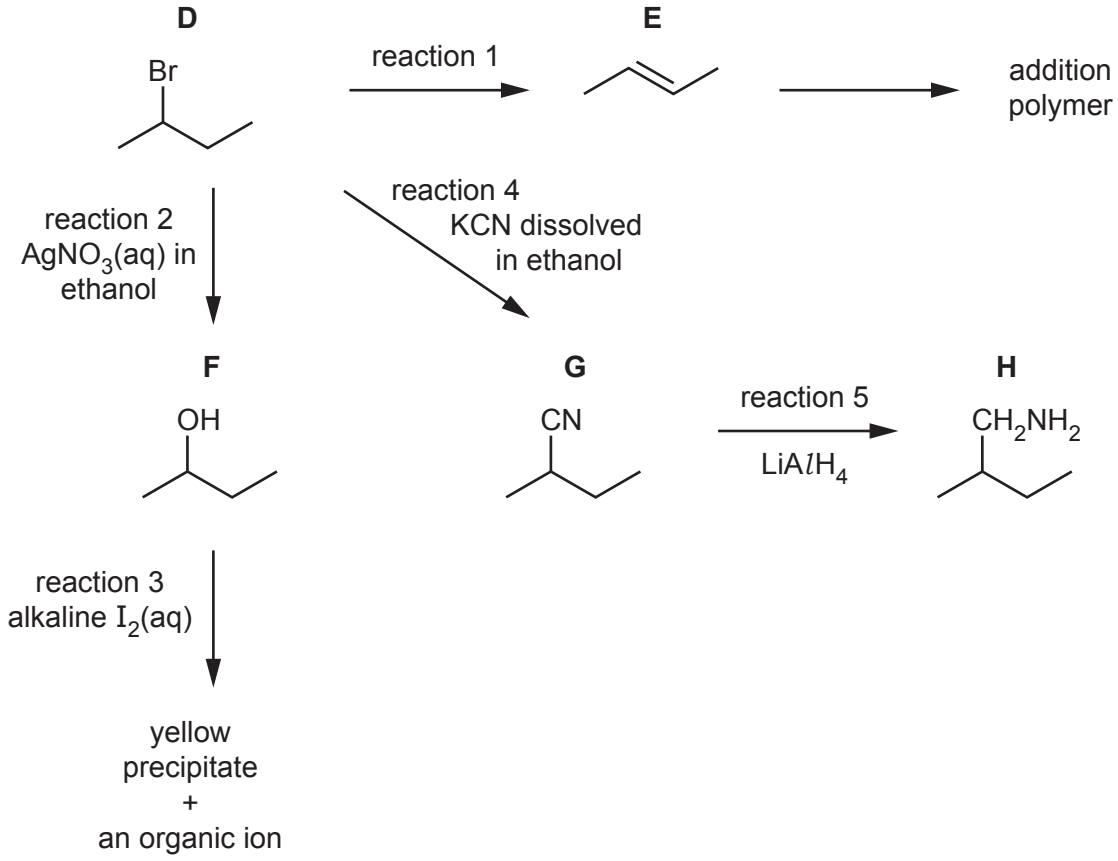


Fig. 4.1

(a) (i) State the reagent and conditions used to form **E** in reaction 1.

..... [1]

(c) (i) State the type of reaction that occurs in reaction 4.

..... [1]

(ii) Reaction 5 is similar to the reaction of $LiAlH_4$ with carboxylic acids to form alcohols.

Suggest the role of $LiAlH_4$ in reaction 5.

..... [1]